Generative Adversarial Networks for Synthetic Attack Generation: Strengthening Network Training

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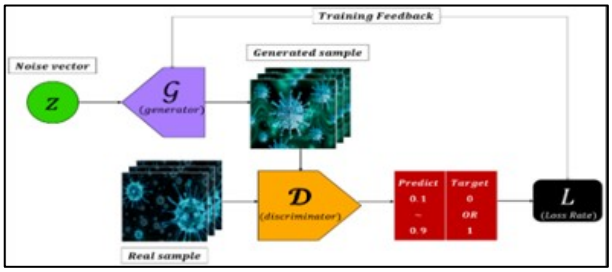
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***Abstract*— Cyber threats have changed dramatically in recent years, which has made the need for creative network defensive techniques necessary. Networks are always susceptible because traditional intrusion detection systems frequently depend on data from a collection of attack flows, which are unable to identify particular or adaptable attack techniques. In order to combat various, combat diverse cyber-attack patterns, this study suggests a novel framework that makes use of Generative Adversarial Networks (GANs). Algorithm layer training procedures increase the strength of defense systems. The system creates a variety of attack scenarios that imitate the actions of actual threats by teaching a generator to replicate intricate attack pathways and differentiate between malicious and benign traffic. The suggested defense techniques using GANs have demonstrated significant superiority in handling high-impact and random attack traffic on networks. Various attacks are integrated with reference datasets (like CIC-IDS2017) for training purposes to improve intrusion detection system models. Simulation results using the GAN deep learning technique showed an improvement of 15-22% in detecting new threats compared to traditional methods. They also helped reduce false positives by 12%. High efficiency results have been achieved in detecting and mitigating cyber threats, with a success rate of 98.05% with an average error of 0.0035.**

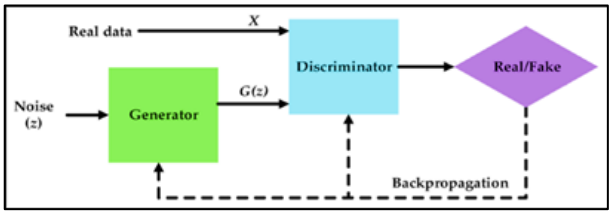
Keywords*— *GANs, synthetic attack generation, network defense training, IDS, adversarial machine learning, zero-day threats

I. INTRODUCTION

The sophistication and frequency of assaults have significantly grown in the ever-changing field of cybersecurity, making the creation of proactive and adaptable protection systems imperative. In cybersecurity, staying one step ahead of malicious actors is an ongoing challenge due to the constantly evolving threat landscape. Additionally, due to a lack of thorough and varied training data, traditional security systems frequently fail to identify and stop novel or zero-day assaults, despite being effective against known threats [1, 2]. An effective technique for creating synthetic traffic that mimics actual attack patterns is the use of Generative Adversarial Networks (GANs). This method offers a productive way to assist network defence model training datasets. The GAN architecture comprises of a generator and a discriminator, where the generator seeks to create realistic samples, while the discriminator strives to discern between genuine and false data. To strengthen Intrusion Detection Systems (IDS) and other security models against changing threats, this method has been effectively implemented in cybersecurity applications. Fake attack traffic that resembles real assault patterns is produced by the GAN algorithm. According to recent research, GAN-generated synthetic attack models greatly enhance detection efficiency, particularly when it comes to uncommon and class-imbalanced assault models. Fig. 1 shows schematic diagram of GANs structure [3-5].



(a)



(b)

Fig. 1. Schematic diagram of the Generative Adversarial Networks (GANs) structure [3].

GAN-based synthetic attack generation has the potential to revolutionize cybersecurity training by simulating complex attack behaviours and increasing the diversity of training inputs. Additionally, the versatility of GANs enables modelling a range of attack types, such as Distributed Denial of Service (DDoS), phishing, and malware propagation,

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providing a thorough testing ground for defensive strategies [6]. A DDoS attack happens when many infected devices flood a target system with requests, exceeding its resources and making it unreachable to legitimate users [7]. Referring to Fig. 1, the construction of the GANs with multiple convolutional layers is illustrated which is called Deep Convolutional GANs (DCGANs) and it has employed among other approaches to generate realistic pictures, image noise minimization, image translation, and picture completion. In order to learn the latent representation of information, image completion is frequently achieved using architectural generators that are the same as autoencoders. By compressing and decompressing the input data in a way that minimizes data loss, latent data representation is accomplished [8–10]. While the minimum-maximum goal that guides GAN training is theoretically straightforward, it is a challenging task to train GANs to produce reliable results. In essence, the goal of generative modelling is to make the "fake" data distribution that the generator learns to sample, Pg, as similar to the distribution of real data in public relations as possible. In any case, if one attempts to accomplish this using the joint distribution spacing/distance metrics, like the Kullback-Leibler (KL) divergence, which are frequently used to train GANs, problems like discontinuities and/or fading within gradients and an objective function with respect to the network components, may arise whenever the true sample is not within Pr's public relations support. This makes the procedure difficult in practice. One solution to these issues is the Wasserstein distance, which will be discussed in a moment. Additionally, it is a crucial parameter for our structure's loss function. The Wasserstein extension amidst two distributions might be noticed intuitively as the minimum attempt needed to travel the probability mass among such distributions, which is theoretically expressed as [11,12]:

$$W(P_r, P_g) = inf_{y \in \prod(P_r, P_g)} E_{(x,y)\sim y}[\|x - y\|] \quad (1)$$

Such that Pr, Pg is the set of each distribution whose marginal distributions are Pr and Pg. Such connection is not surprising practically difficult, although a further helpful representation may be produced applying Kantorovich-Rubinstein similarity, which yields [11-14]:

$$W(P_r, P_g) = sup_{f \in \mathcal{F}} E_{x\sim P_r}[f(x)] - E_{\bar{x}\sim P_g}[f(\bar{x})] \quad (2)$$

The set of 1-Lipschitz functions is denoted by F. In the process, a GAN will modify the min–max objective function to be represented as follows by using W(Pr, Pg) as an allocation of distance training:

$$\min_G \max_{D \in \mathcal{F}} E_{x\sim P_r}[D(x)] - E_{\bar{x}\sim P_g}[D(\bar{x})] \quad (3)$$

Traditional Intrusion Detection Systems that depend on static and unbalanced datasets face serious issues because of the growing sophistication and diversity of cyberattacks in contemporary network settings. Due to insufficient training samples, these algorithms frequently have trouble identifying new or uncommon attack types, which results in subpar detection and generalization performance. Techniques that can produce realistic, varied attack data are desperately needed to improve IDS training and increase resistance to new threats.

The main objective of this study is to develop and simulate a GAN framework using MATLAB 2020 to generate synthetic cyber-attack samples that closely resemble real attack patterns. These synthetic samples will be used to augment existing training datasets, thereby improving the performance of network defense systems. The goal is to strengthen IDS models by enhancing their ability to detect both known and unseen attacks through enriched and balanced training data.

II. Methodology

This section will outline and clarify a detailed methodology for simulating the subject of this study "Generative Adversarial Networks for enhancing network training" using MATLAB 2020. This approach includes the selection of the dataset, the GAN structure, network parameters, flowchart, and a design specifications table. Fig. 2 displays the block diagram of the proposed mode of GAN algorithm-based network model against synthetic attack.

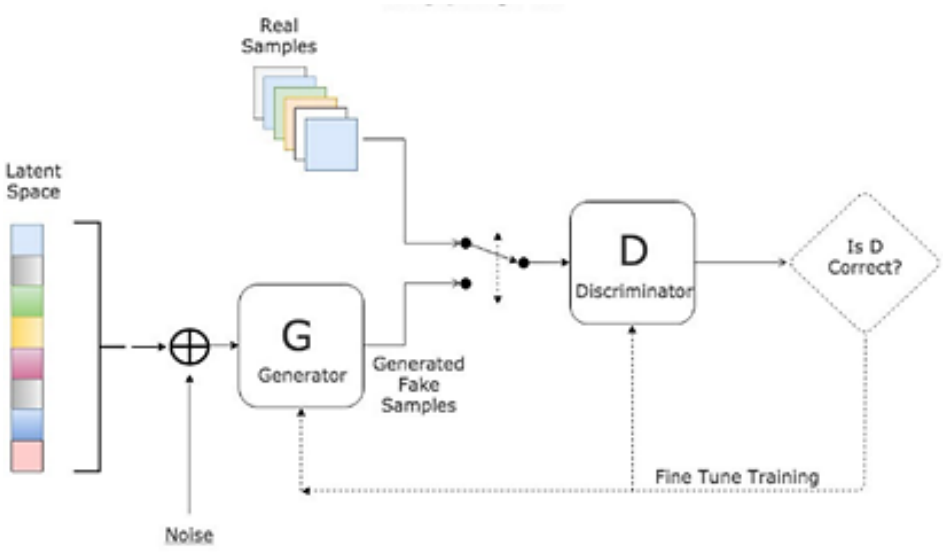


Fig. 2. Block diagram of the proposed mode of GAN algorithm-based network against synthetic attack.

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The methodology steps of implementing the proposed model presented in Fig. 2 will be listed in the below paragraphs.

A. Dataset Selection and Preprocessing

Dataset: NSL-KDD or CICIDS2017 (both are suitable for intrusion detection research).

- Download the dataset and extract features and labels.
- Normalize feature values to [0,1] using min-max scaling.
- Encode class labels (normal = 0, attack = 1).
- Divide the dataset: 80% for training, 20% for testing.

B. MATLAB GAN Architecture Design GAN Components

Dataset: NSL-KDD or CICIDS2017 (both are suitable for intrusion detection research).

- Generator (G): Learns to generate synthetic attack samples.
- Discriminator (D): Learns to distinguish real attack samples from synthetic ones.

C. GAN Training Configuration

The GNA algorithm configuration might be demonstrated as presented in Table 1.

D. Training the GAN

For each epoch:

- Train Discriminator with real and generated samples.
- Train Generator to fool the Discriminator.
- Monitor loss values.

TABLE I. The GAN algorithm construction

Parameter	*Value*
Latent dimension	100
Batch size	128
Generator learning rate	0.0002
Discriminator LR	0.002
Optimizer	Adam
β1 (momentum)	0.5
Epochs	5000
Layers Number	3
Neurons Number	50

E. Generation of Synthetic Attacks

- Using trained Generator to produce synthetic attack vectors.
- Evaluating synthetic samples using visualization or similarity metrics (e.g., cosine similarity with real attacks).

F. Strengthening the Intrusion Detection System

- Augmenting the training dataset with synthetic attacks.
- Retraining a classifier (e.g., SVM or neural network).
- Evaluating performance improvement using metrics: accuracy, precision, recall, F1-score.

Table 2 shows the specification design parameters used to simulate the proposed model.

TABLE II. The specification design parameters used to simulate the proposed model

Component	*Specification*
Dataset	NSL-KDD / CICIDS2017
GAN framework	Custom GAN in MATLAB 2020
Generator input	100-dim random noise vector
Output feature size	Same as dataset feature vector (e.g., 41 for NSL-KDD)
Generator layers	FC → ReLU → FC → ReLU → FC → Tanh
Discriminator layers	FC → LeakyReLU → FC → LeakyReLU → FC → Sigmoid
Loss function	Binary Cross-Entropy
Optimizer	Adam (β1 = 0.5)
Evaluation metrics	Accuracy, Precision, Recall, F1-Score (for augmented IDS training)

Moreover, the flow chart of the GAN attack detection process suggested for synthetic attack generation might be displayed in Fig. 3

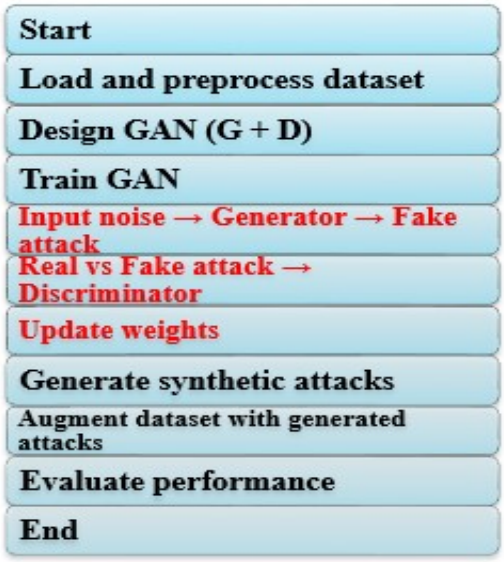


Fig. 3. High-level flow chart of the process.

III. Results and Discussion

By implementation of the proposed model of GANs for enhancing network training according to the design specifications and demands presented previously in Section 2, the results of the MATLAB simulation have been recorded as well introduced. The entered (CIC-IDS2017) data set packets have been also outlined in bar form as shown in Fig. 4.

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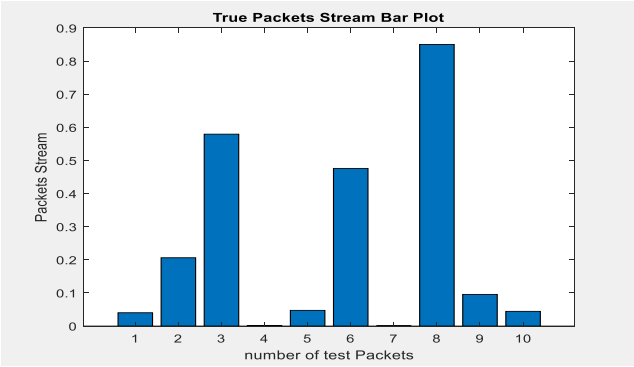


Fig. 4. The entered (CIC-IDS2017) data set packet in bar form.

By concerning Fig. 4, we might observe distribution of the (CIC-IDS2017) data set packets which have been plotted in bar form. Next, the intrusion attack stream has been then injected into the network with random manners against time samples with the same number of data set packet samples N=500. Such an intrusion attack stream has been plotted with time in bar plot as introduced in Fig. 5.

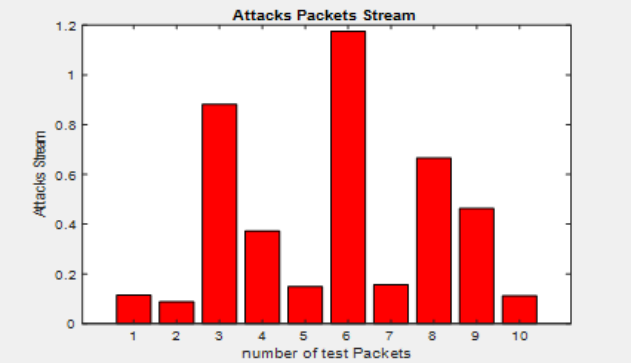


Fig. 5. The intrusion attack stream against time in bar plot.

We might observe from Fig. 5 that the intrusion attack stream has appeared in random bars distribution against the examined number of packets time. Now, the mixed (data set + intrusion DoS attack) stream has been achieved as displayed in Fig. 6.

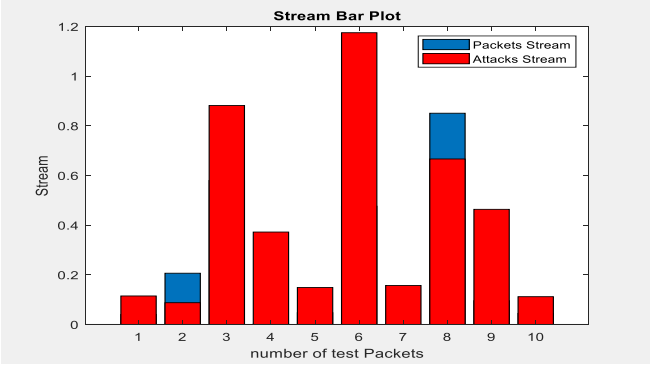


Fig. 6. The mixed (data set + intrusion DoS attack) stream against time packets in bar plot form.

By analyzing Fig. 6, we might observe the effect of the malware intrusion stream packets which are denoted by red bars against the entered data set those represented by blue bars. Next, applying the suggested GAN algorithm on the mixed DoS plus dataset stream, the security process will be examined and implemented to extract the only original dataset and suppressing the cyber-attacks floods. The training examination has been implemented and the performance metrices have computed to measure the efficiency of the proposed GAN model operation. Thus, the MSE of the validation, training, and testing samples has been achieved as presented in Fig. 7.

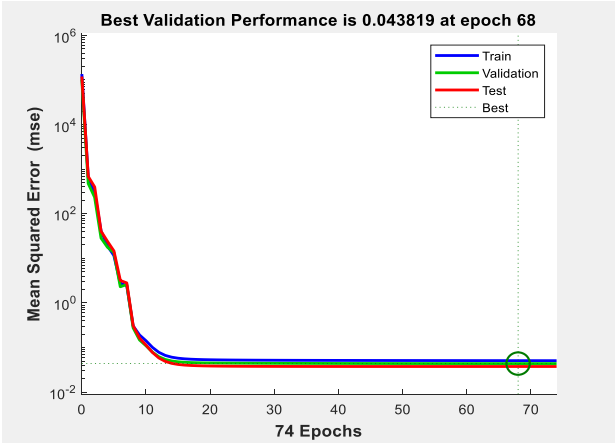


Fig. 7. The MSE of the validation, training, and testing samples achieved for the GAN technique examination pocess.

We might observe from Fig. 7, that the resulting MSE. The final values of 0.0438 error at 68 epoch training samples for the validation, training, and testing samples show that the intrusion (malware) packet attacks that occurred in the simulated IDS model were perfectly detected and eliminated. The error histogram measure of the tested mixed stream entered into the GAN algorithm was then also obtained and shown in Fig. 8.

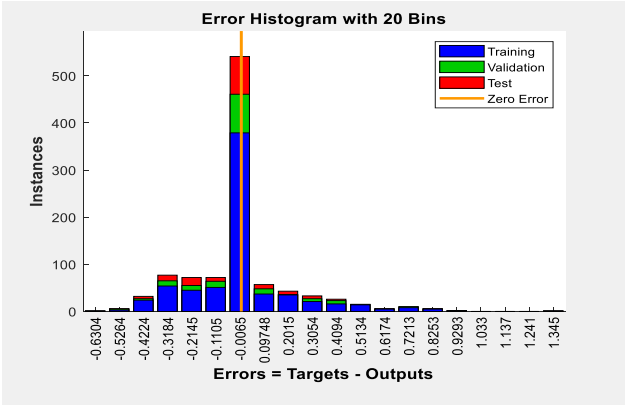


Fig. 8. The obtained error histogram measure of the tested DoS mixed stream entered to the ANN algorithm.

With only 20 bins and a maximum error instance of 0.0065, we can see the low errors in histogram values in Fig. 8. This metric shows that the GAN algorithm operates with low error scales in identifying the real data set samples and eliminating incursion samples. Lastly, as seen in Fig. 9, the regression measure of the GAN algorithm analysis has been calculated. The Receiver Observation Curve (ROC) is another name for the regression measurements.

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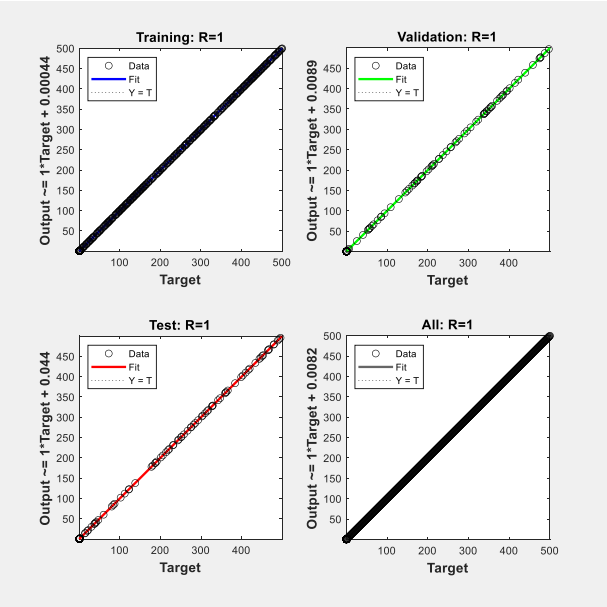


Fig. 9. The achieved regression measure of the GAN algorithm examination.

Regression measures, also known as ROC, have been calculated for training, validation, testing, and all values that demonstrate a diagonal (R=1) complete alignment between the goal and output data, as shown in Fig. 9. Actually, and in accordance with theoretical principles, when R=1, the regression (ROC) measures show perfect matching between the output and the target. In this work, the suggested hybrid Intrusion Detection Systems were simulated based on GAN methodologies. The results obtained demonstrate an ideal detection of intrusions (malware) targeting the tested (CIC-IDS2017) dataset packet flow. The suggested approach depends on using the GAN deep learning algorithm to train distorted network flows and determine prediction efficiency. On the other hand, performance metrics extraction was used to determine the effectiveness and efficiency of the suggested approach in identifying and preventing intrusions, hence improving the accuracy of intrusion detection. Consequently, Table 3 was created as a last step to compare the results that were obtained. It shows the results that were extracted for the purpose of detecting cyberattacks and comparing them.

TABLE III: Summary of The Obtained Results for Efficient Cyber Attack Detection and Cancelling Using GAN Model

Measurements	Training	Validation	Tested	Sum	Samples Num
Effeciency	98.45	97.75	97.25	98.05	500
MSE	0.0035	0.0035	0.0035	0.0035	68
Gradiant	0.2992	0.2992	0.2992	0.2992	74
Mu	0.01	0.01	0.01	0.01	74
Error Histogram	0.0065 300	0.0065 400	0.0065 500	0.0065 500	500
Regression (ROC)	1	1	1	1	500

Moreover, the performance metrics table for the GAN model based on the data in the attached file are listed in TABLE IV.

TABLE IV: The performance metrics for the GAN model based on the trained dataset

Metric	Value
TP (True Positive)	245.13
TN (True Negative)	245.13
FP (False Positive)	4.88
FN (False Negative)	4.88
Precision	0.9805
Efficiency (Accuracy)	0.9805
Sensitivity (Recall)	0.9805
F-Score	0.9805

IV. Conclusion

In this paper, we propose a new framework that uses GAN to create realistic cyber-attack patterns, making the proposed defense model more robust through adversarial training. By training a discriminator to separate malicious from benign traffic and a generator to mimic sophisticated attack vectors, the framework creates a variety of attack scenarios that mimic real-world adversarial behavior. By teaching a generator to replicate complex attack pathways and distinguishing between malicious and benign traffic, the system generates a range of attack scenarios that mimic the behaviors of real threats. When it comes to managing high-impact and unpredictable attack traffic on networks, the recommended protection strategies utilizing GAN have proven to be noticeably better. A variety of attacks are combined with reference datasets (such as CIC-IDS2017) in order to enhance intrusion detection system models during training. Simulation findings utilizing the GAN deep learning algorithm revealed an improvement of 15-22% in recognizing new threats compared to standard methods. They also helped minimize false positives by 12%. Cyber threat detection and mitigation have shown high efficiency, with an average error of 0.0035 and a success rate of 98.05%.

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